



User Guide for the DEN Paper Template

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Abstract

An abstract summarizes, in one paragraph of 300 words or fewer, the major aspects of the entire paper. It should include: 1) the overall purpose of the study and the research problem you investigated; 2) the basic design of your research approach; 3) major findings as a result of your analysis; and 4) a brief summary of your interpretations and conclusions.

Keywords: keyword; keyword-two; keyword number three

JEL Classification: C22, O47

1. Introduction

1.1 Background

Trade liberalization has long been promoted as a pathway to economic growth. The classical theories of Ricardo and Heckscher-Ohlin predict that reducing tariff barriers allows countries to specialize in their comparative advantage, lowering consumer prices and raising aggregate output. International institutions, including the World Bank, the IMF, and the WTO, have consistently advocated trade openness as a pillar of development strategy, and Indonesia has followed this approach through successive rounds of tariff reduction. From WTO accession in 1995 to AFTA, ACFTA, RCEP, and a series of bilateral agreements, Indonesia's simple average applied tariff has fallen from approximately 22 percent in the late 1980s to 5.99 percent in 2023, while the trade-weighted average stood at 2.07 percent in the same year (World Bank), as shown in Figure 1.

Manufacturing has long been seen as an important driver of economic growth and a way for countries to escape the middle-income trap, a term introduced by Gill and Kharas (2007) to describe economies that stop growing after reaching middle-income status. The experiences of South Korea, Taiwan, and Singapore illustrate this process, as these economies moved workers from low-productivity activities into higher-value manufacturing. From Kaldor (1967) to Rodrik (2016), researchers have identified manufacturing as an important sector in supporting this transition. Manufacturing can provide jobs for low-skilled workers while offering higher productivity, encourage learning and the spread of new technologies, and produce tradable goods that generate foreign exchange. These features are closely related to the increasing returns in industrial production described by Kaldor (1967), as well as the finding by Rodrik (2013) that labour productivity in formal manufacturing tends to converge across countries, unlike other sectors of the economy.

In Indonesia, manufacturing contributed 1.07 percentage points to the country's 5.11 percent GDP growth in 2025, exceeding the contributions of agriculture, trade, mining, and services (BPS, 2026). However, Indonesia's manufacturing sector has been declining in relative importance. Its share of GDP fell from 27.8 percent in 2008 to 18.3 percent in 2022, placing Indonesia among the ASEAN countries with the lowest manufacturing shares. This decline reflects a pattern of premature deindustrialization, in which manufacturing begins to lose its role as an engine of growth before the economy reaches a high-income level. If manufacturing is the ladder out of the middle-income trap, Indonesia's ladder appears to be shrinking before the country has finished climbing. Therefore, it is important to analyze how trade liberalization affects the manufacturing sector.

Beyond its implications for the manufacturing sector as a whole, an important question is whether the benefits of trade liberalization are broadly shared or instead concentrated among certain firms and workers. Melitz (2003) demonstrates that trade liberalization favors large, productive, export-oriented firms, which gain access to cheaper inputs and larger markets. Small, domestic-market firms, by contrast, face the downside of increased import competition without offsetting export opportunities. In Indonesia, this distributional concern is particularly acute because the manufacturing sector is dominated not by large factories but by millions of micro and small enterprises. Xxx (here I will add the data of wage workers in Indonesia from Susenas).

In the framework of Lewis (1954), unpaid family workers represent the low-productivity subsistence sector, while paid workers represent the higher-productivity modern sector. Structural transformation, in the Lewisian sense, occurs when workers transition from subsistence to modern employment. The central question of this study is therefore whether trade liberalization facilitates or impedes this transition for Indonesia's micro and small manufacturers.

The theoretical literature on trade and informality suggests that the direction of this effect is ambiguous. Paz (2014), building on a Melitz-type heterogeneous-firm model with endogenous formality decisions, shows that domestic tariff cuts can either increase or decrease the informal employment share depending on initial labor market conditions. Goldberg and Pavcnik (2003) provide empirical evidence from Colombia and Brazil that tariff reductions increased informality in manufacturing. More recently, Dix-Carneiro and Kovak (2019) demonstrate that the informality effects of trade liberalization in Brazil persisted for nearly two decades. In contrast, McCaig and Pavcnik (2018) find that export market access in Vietnam pulled workers from informal household enterprises into formal manufacturing. The net effect therefore depends on whether import competition, which may favor informality, or export access, which may favor formality, dominates.

The continuous decline in tariffs does not necessarily mean that trade has become unrestricted. Governments have relied increasingly on non-tariff measures (NTMs), defined as policy measures other than ordinary customs tariffs that can have an economic effect on international trade in goods by changing the quantities traded, the prices, or both (UNCTAD, 2019). NTMs include sanitary and phytosanitary requirements, technical barriers to trade, import licensing and approval requirements, pre-shipment inspection, and restrictions on the port of entry. It is important to note that this definition is deliberately neutral. A regulation is classified as an NTM whenever it has the potential to affect the price or quantity of traded goods, and the classification itself makes no judgement about whether the measure is legitimate or discriminatory (Doan and Rosenow, 2019). A high count of NTMs is therefore not in itself evidence of protectionism. As incomes rise, so do public demands for health, safety, and environmental standards, and some NTMs can expand trade by reducing information asymmetries and signalling product quality. The relevant question is not how many NTMs a country maintains but what they cost the firms that must comply with them.

1.2 Research Question

1.2.1 To what extent has trade liberalization, measured by the Non-tariff Measure, affected the share of unpaid workers in Indonesian big and medium manufacturing industries during xxx?

2. Studi Literatur

2.1 Non-Tariff Measure

Non-tariff measures are trade policy instruments other than tariffs that have the potential to affect international trade in goods, whether by altering traded quantities, prices, or both. Their importance has grown as global tariff levels have fallen, a decline driven by successive rounds of negotiation under the General Agreement on Tariffs and Trade and its successor, the World Trade Organization. Rising consumer concern over safety, food quality, and environmental protection has reinforced this shift, as governments have turned to regulatory instruments to pursue objectives that tariffs cannot address (CIPS 2025).

UNCTAD (2019) classifies non-tariff measures into sixteen chapters. Import measures are divided into technical and non-technical categories. The technical measures comprise sanitary and phytosanitary measures (A), technical barriers to trade (B), and pre-shipment inspection and other formalities (C). The non-technical measures cover contingent trade-protective measures (D), non-automatic import licensing, quotas, prohibitions, and other quantity-control measures (E), price-control measures including additional taxes and charges (F), finance measures (G), measures affecting competition (H), trade-related investment measures (I), distribution restrictions (J), restrictions on post-sales services (K), subsidies and other forms of support (L), government procurement restrictions (M), intellectual property (N), and rules of origin (O). A final chapter covers export-related measures (P).

The trade costs generated by these measures now exceed those generated by tariffs in the majority of cases. According to the most recent UNCTAD assessment, non-tariff measures impose a larger cost burden than tariffs for 88 percent of countries, and developing economies are doubly disadvantaged in facing both higher tariffs and higher compliance costs (UNCTAD 2026).

Indonesia illustrates this pattern. Between 2015 and 2018 the number of regulations containing non-tariff measures rose by close to 14 percent, from 169 to 192. The count of coded measures grew more sharply over the same period, from 767 to 977, and these 977 coded measures applied to 7,759 tariff lines in total (ERIA 2019).

2.2 Trade Policies on Labor

A growing body of research has examined the labor-market effects of trade liberalization, with much of the evidence focusing on tariff reductions. These studies show that the effects of tariff cuts on labor-market outcomes depend on the direction of liberalization, the characteristics of affected firms and workers, and the institutional context.

Kis-Katos and Sparrow (2015) combine the annual manufacturing survey (Survei Industri) with Sakernas for the period 1993 to 2002 and estimate the effects at both the provincial and district levels in Indonesia. Both levels of analysis indicate that average wages rose in response to reductions in tariffs on intermediate inputs. The two types of tariffs also affect poverty in opposite directions, since lower output tariffs are associated with higher poverty across the poverty measures, while lower input tariffs are associated with poverty reduction.

Amiti and Konings (2007) find that a 10-percentage-point fall in input tariffs raises productivity by about 12 percent for firms that import their inputs, at least twice the gain from an equivalent cut in output tariffs. To be precise, importing firms receive this full benefit, while non-importers receive only about 3 percent, indicating that the gains from input liberalization are concentrated among a subset of firms rather than shared evenly across the sector.

Evidence from Brazil points to a further source of heterogeneity, this time across labor-market segments rather than across firms. Paz (2014) finds that the gains accrued disproportionately to formal employment. A one-percentage-point reduction in the tariffs applied by trading partners to Brazilian exports raised formal-sector wages by roughly 0.32 percent, whereas a one-percentage-point reduction in Brazil's own import tariffs lowered formal wages by about 0.05 percent. Lower partner tariffs also reduced the probability that a worker was employed informally.

The evidence is not uniformly positive. Delesalle *et al.* (2025) find no significant average effect of the ASEAN-China Free Trade Area on the probability of holding a wage job in Indonesia. The estimated overall effect is economically very small or statistically insignificant for men, women, and less-educated workers. It is larger and statistically significant only for educated workers, for whom the average increase is 1.9 percentage points against a baseline probability of 17.4 percent in 2000. Since this group accounts for around a quarter of the workforce, the distributional incidence of the agreement was regressive. Topalova (2007) reports a sharper result for India, where a district exposed to the average tariff reduction recorded a poverty headcount roughly two percentage points higher and a poverty gap roughly 0.6 percentage points higher than a district experiencing no tariff reduction.

The effects also depend on the institutional setting. Goldberg and Pavcnik (2003) find that Colombian tariff reductions raised informal employment by roughly 0.1 percentage points per percentage point of tariff cut, but only before Law 50 of 1990 reduced firing costs. Post-reform, the interaction between tariffs and informal employment is consistently positive, indicating that the same tariff decline leads to a smaller increase in informality afterwards, and possibly a decrease. These results indicate that trade policy affects informality only when formal employment is costly to adjust.

Goldberg and Pavcnik (2007) find that globalization increases the wage premium for skilled workers in developing countries at the economy-wide level, within industries, and within firms. Amiti and Cameron (2012), however, report the opposite for Indonesian manufacturing. Using firm-level data with firm and year-by-island fixed effects, they find that a 10-percentage-point reduction in input tariffs lowered the within-firm skill premium by about 1.4 to 2.6 percent, while output tariffs had no significant effect once input tariffs were controlled for. The effect is concentrated among firms that actually import intermediate inputs.

##Empirical Studies

Trade liberalization and welfare has been the subject of a number of empirical research. Standard trade models predict aggregate welfare gains from the removal of trade barriers, where liberalization allows economies to specialise according to comparative advantage and open consumers to access of more varied goods (Krugman, Obstfeld and Melitz, 2022). These aggregate predictions, however, say little about how the gains are distributed, and the empirical literature has increasingly turned to that question.

Much of the recent evidence shows that the direction of the effect depends on which tariffs are cut. Kis-Katos and Sparrow (2015) combine the annual manufacturing survey (Survei Industri or IBS) with Sakernas for the period 1993 to 2002 and estimate effects at both the provincial and the district level in Indonesia. Both levels of analysis indicate that average wages rose in response to reductions in tariffs on intermediate inputs. The two tariff types also move poverty in opposite directions, since lower output tariffs are associated with higher poverty across the poverty measures while lower input tariffs are associated with poverty reduction.

Amity and Konings (2007) find that a 10 percentage point fall in input tariffs raises productivity by about 12 percent for firms that import their inputs, at least twice the gain from an equivalent cut in output tariffs. To be precise, importers receive this full benefit while non-importers receive only about 3 percent, so the gains from input liberalization are concentrated among a subset of firms rather than

shared evenly across the sector.

Evidence from Brazil points to a further source of heterogeneity, this time across labour market segments. Paz (2014) finds that the gains accrued disproportionately to formal employment. A one percentage point reduction in the tariffs applied by trading partners to Brazilian exports raised formal-sector wages by roughly 0.32 percent, whereas a one percentage point reduction in Brazil's own import tariffs lowered formal wages by about 0.05 percent. Lower partner tariffs also reduced the probability that a worker was employed informally.

The evidence is not uniformly positive. Delesalle et al. (2025) find no significant average effect of the ASEAN-China Free Trade Area on the probability of holding a wage job in Indonesia. The estimated overall effect is economically very small or statistically insignificant for men, for women, and for less-educated workers. It is larger and statistically significant only for educated workers, for whom the average increase is 1.9 percentage points against a baseline probability of 17.4 percent in 2000. Since this group accounts for around a quarter of the workforce, the distributional incidence of the agreement was regressive. Topalova (2007) reports a sharper result for India, where a district exposed to the average tariff reduction recorded a poverty headcount roughly two percentage points higher and a poverty gap roughly 0.6 percentage points higher than a district whose tariffs were unchanged.

Whether such adverse effects materialise appears to depend on the institutional setting. Goldberg and Pavcnik (2003) find that Colombian tariff reductions raised informal employment by roughly 0.1 percentage points per percentage point of tariff cut, but only before Law 50 of 1990 reduced firing costs. Post-reform, the interaction between tariffs and informal employment is consistently positive, indicating that the same tariff decline leads to a smaller increase in informality afterwards, and possibly a decrease. These results indicate that trade policy moves informality only when formal employment is costly to adjust.

Goldberg and Pavcnik (2007) find that globalization increases the wage skill premium in developing countries at the economy-wide level, within industries, and within firms. Amiti and Cameron (2012), however, report the opposite for Indonesian manufacturing. Using firm-level data with firm and year by island fixed effects, they find that a 10 percentage point reduction in input tariffs lowered the within-firm skill premium by about 1.4 to 2.6 percent, while output tariffs had no significant effect once input tariffs were controlled for. The effect is concentrated among firms that actually import intermediates.

In analysing the effects of trade liberalization on poverty and labour market outcomes, Kis-Katos and Sparrow (2015) construct district-level tariff exposure by combining nationally determined sectoral import tariffs with each district's initial economic structure prior to trade liberalization. This approach follows a **shift-share (Bartik) framework**, in which national tariff changes are weighted by predetermined district-level employment or industrial output shares.

For the poverty analysis, district output tariff exposure is measured using the initial employment distribution across sectors. Specifically, the tariff exposure of district k in year t is defined as

$$Tariff_{kt} = \sum_{s=1}^S \left(\frac{Q_{sk,1990}}{Q_{k,1990}} \times Tariff_{st} \right),$$

where:

- $Q_{sk,1990}$ denotes employment in output sector s in district k in 1990.
- $Q_{k,1990}$ is the total labour force in district k in 1990.
- $Tariff_{st}$ is the national import tariff applied to sector s in year t .
- S denotes the total number of tradable sectors.

For the firm-level analysis, Kis-Katos and Sparrow (2015) alternatively construct an output tariff measure using the initial industrial structure of each district. In this case, sectoral tariffs are weighted by each sector's share in the district's total industrial output in 1993, $\left(\frac{Q_{sk,1993}}{Q_{k,1993}}\right)$, rather than by employment shares.

To capture the effects of tariffs on intermediate inputs, the authors also construct an input tariff measure. The input tariff for each industry is first calculated as the weighted average tariff on all imported intermediate inputs, where the weights are given by each input's share in the industry's total input use. These industry-specific input tariffs are then aggregated to the district level using the district's initial employment shares:

$$InputTariff_{kt} = \sum_{s=1}^S \left(\frac{Q_{sk,1990}}{Q_{k,1990}} \times \sum_{j=1}^J \left(\frac{M_{js,1990}}{M_{s,1990}} \times Tariff_{jt} \right) \right).$$

where:

- $M_{js,1990}$ denotes the value of intermediate input j used by industry s in 1990.
- $M_{s,1990}$ is the total value of intermediate inputs used by industry s in 1990.
- $\frac{M_{js,1990}}{M_{s,1990}}$ represents the share of input j in the total intermediate inputs of industry s .
- $Tariff_{jt}$ denotes the national tariff on intermediate input j in year t .
- J represents the total number of input sectors.

Consistent with Kovak (2013), Kis-Katos and Sparrow (2015) exclude non-tradable sectors from the construction of the output tariff measure because these sectors are not directly exposed to international trade.

The effects of trade liberalization on poverty and labour market outcomes are then estimated using the following first-difference specification:

$$\Delta y_{kt} = \alpha + \beta_1 \Delta OutputTariff_{kt} + \beta_2 \Delta InputTariff_{kt} + \Delta X'_{kt} \gamma + I'_k \theta + \lambda_{rt} + \Delta \varepsilon_{kt},$$

where:

- Δy_{kt} denotes the change in the district-level outcome variable, including measures of poverty, employment, and wages.
- $\Delta OutputTariff_{kt}$ represents the change in district-level output tariff exposure.
- $\Delta InputTariff_{kt}$ represents the change in district-level input tariff exposure.
- ΔX_{kt} is a vector of time-varying control variables, including the share of the rural population, the share of the working-age population (16–60 years), adult literacy rates, and minimum wages.
- I_k is a vector of initial district characteristics, including the 1990 labour shares used as tariff weights, aggregated to one-digit sectors, the 1990 rural population share, and, in some specifications, the initial level of the dependent variable.
- λ_{rt} denotes island-by-year fixed effects.
- $\Delta \varepsilon_{kt}$ is the error term.

The first-difference specification removes time-invariant district characteristics and exploits within-district variation in tariff exposure over time. By combining predetermined employment shares with nationally determined tariff changes, the Bartik framework identifies heterogeneous district exposure to trade liberalization while reducing concerns regarding reverse causality.

Unlike Kis-Katos and Sparrow (2015), who estimate the model using a first-difference OLS specification, Delesalle et al. (2025) employ a Bartik instrumental variable (IV) strategy to address the potential endogeneity of tariff reductions. District-level exposure to trade liberalization depends on both the pre-liberalization industrial structure and industry-level tariff changes. Rather than examining multilateral trade liberalization, the authors focus exclusively on the bilateral tariff reductions between Indonesia and China under the ASEAN–China Free Trade Agreement (ACFTA), using pre-ACFTA industry shares to construct the exposure measures. Specifically, they consider three channels through which trade liberalization may affect local labor markets: output import tariff cuts, input import tariff cuts, and export tariff cuts.

The output import tariff measure is constructed as

$$\Delta OutputImportTariff_{d,2000-2010} = \sum_s \frac{L_{s,d,2000}}{L_{d,2000}} \cdot \Delta Tariff_{s,2000-2010}^{Indo}$$

where:

- $L_{s,d,2000}$ is the number of working-age individuals employed in industry s in district d in 2000.
- $L_{d,2000}$ is the total working-age population employed in district d in 2000.
- $\Delta Tariff_{s,2000-2010}^{Indo}$ denotes the change in Indonesia's import tariff on Chinese products in industry s between 2000 and 2010.

To capture the effect of imported intermediate inputs, the authors construct the input import tariff measure:

$$\Delta InputImportTariff_{d,2000-2010} = \sum_s \frac{L_{s,d,2000}}{L_{d,2000}} \left(\sum_j \frac{M_{j,s,2000}}{M_{s,2000}} \cdot \Delta Tariff_{j,2000-2010}^{Indo} \right)$$

where:

- $M_{j,s,2000}$ denotes the value of intermediate input j used by industry s .
- $M_{s,2000}$ is the total intermediate inputs used by industry s .
- $\frac{M_{j,s,2000}}{M_{s,2000}}$ represents the cost share of input j in industry s .
- The inner summation calculates the industry-level input tariff, while the outer summation aggregates these tariffs to the district level using initial employment shares.

The export tariff measure follows the same specification as the output import tariff but replaces Indonesian import tariffs with Chinese tariffs imposed on Indonesian exports:

$$\Delta ExportTariff_{d,2000-2010} = \sum_s \frac{L_{s,d,2000}}{L_{d,2000}} \cdot \Delta Tariff_{s,2000-2010}^{China}$$

where $\Delta Tariff_{s,2000-2010}^{China}$ denotes the change in Chinese import tariffs on Indonesian exports in industry s .

The relationship between tariff exposure and labor market outcomes is estimated using the following empirical specification:

$$\Delta y_{d,2000-2010} = \alpha + \beta_1 \Delta OutputImportTariff_d + \beta_2 \Delta InputImportTariff_d + \beta_3 \Delta ExportTariff_d + \gamma X_{d,2000}$$

where:

- $\Delta y_{d,2000-2010}$ denotes the change in the district-level labor market outcome between 2000 and 2010.
- $\Delta OutputImportTariff_d$ is the district-level output import tariff exposure.
- $\Delta InputImportTariff_d$ is the district-level input import tariff exposure.
- $\Delta ExportTariff_d$ is the district-level export tariff exposure.
- $X_{d,2000}$ is a vector of pre-liberalization district characteristics, including industrial composition, education, demographic variables, and island fixed effects.
- ε_d is the error term.

To address the potential endogeneity of tariff reductions, Delesalle et al. (2025) instrument district-level tariff changes using the initial average tariff level in 2000. The identification strategy exploits the design of the ACFTA, under which sectors with higher pre-liberalization tariffs experienced systematically larger tariff reductions between 2000 and 2010. Consequently, the initial tariff level provides a strong predictor of subsequent tariff changes while remaining plausibly exogenous to later labor market outcomes after conditioning on pre-treatment characteristics.

Unlike Kis-Katos and Sparrow (2015), who rely on a first-difference OLS specification, Delesalle et al. (2025) estimate the causal effects using a Bartik IV strategy, thereby strengthening causal identification by mitigating potential bias arising from the non-random allocation of tariff reductions across industries.

3. Metodologi

3.1 Data

3.1.1 Data and Unit of Analysis

This study utilizes the Survei Industri Besar dan Sedang (IBS) administered by Badan Pusat Statistik (BPS) for the period 2008 to 2023. The IBS enumerates manufacturing establishments with 20 or more workers, which distinguishes it from the Survei Industri Mikro dan Kecil (IMK), covering establishments with 1 to 19 workers. The 2008 to 2023 window is selected for three reasons. It begins before the most substantial phase of Indonesia's recent trade liberalization, it spans both the pre-Covid-19 and post-Covid-19 periods, and it corresponds to the years for which World Bank non-tariff measure data are available.

An important measurement constraint should be noted at the outset. Both the IMK and the IBS record workers only as paid or unpaid, and this classification does not correspond to informality as defined in the comparative literature. Goldberg and Pavcnik (2003) define informality through compliance with labour market regulation, a definition that neither survey can support because neither collects consistent information on contracts, social security enrolment, or regulatory registration. The IBS questionnaire does contain additional labour items in xxx, including workers engaged under fixed-term contracts (Pekerja dengan Perjanjian Kerja Waktu Tertentu), but these items are not asked consistently across the full period of analysis. The paid and unpaid distinction is therefore the only labour classification that can be measured on a comparable basis in both surveys and across all years. This study accordingly follows the approach of Delesalle et al. (2025), who proxy job quality by wage-employment status rather than by regulatory compliance.

Data on non-tariff measures are drawn from the World Bank Indonesia NTM database, which covers the period 2008 to 2023 at monthly frequency. The database is structured as a panel at the product-month-year level in which individual non-tariff measures, along with tariffs, are recorded as binary indicators. A companion file at the NTM-product-regulation-month-year level records the regulation underlying each measure and the issuing ministry or government agency, and this file forms the basis on which the product-month-year panel is constructed. The regulation-level file contains base regulations,

subsequent amendments, and revocations, together with the product codes and NTM codes associated with each regulation (World Bank 2023).

3.1.2 Distribution Sample

The provincial distribution of IMK firms during the 2018–2022 period reveals substantial variation across Indonesia. West Java recorded the highest number of IMK firms in the sample, with 65 firms, indicating that the province has the greatest concentration of small manufacturing establishments among all provinces included in the dataset. **|Province| Number of IMK Firms|** |:-----|-----:|
 | Aceh | 7 || North Sumatra | 16 || West Sumatra | 4 || Riau | 29 || Jambi | 1 || Lampung | 13 || Riau
 Islands | 5 || DKI Jakarta | 23 || West Java | 65 || Central Java | 35 || DI Yogyakarta | 18 || East Java
 | 61 || Banten | 8 || Bali | 9 || West Nusa Tenggara | 4 || West Kalimantan | 2 || Central Kalimantan |
 4 || East Kalimantan | 10 || North Sulawesi | 10 || Central Sulawesi | 1 || South Sulawesi | 1 || West
 Papua | 8 || Papua | 9 || ## Metode

4. Hasil

5. Kesimpulan

6. Referensi

6.1 References

Add your bibliography entries to `ref.bib` and cite them using standard Quarto citation syntax. Here is an example citation on diamond open access (**fuchs2013diamond?**). According to (**fuchs2013diamond?**), diamond open access model is amazing.

6.2 Footnotes

Use footnotes sparingly. They should be used for additional information that is not essential to the main text.¹

6.3 Tables

Use standard markdown tables or Quarto's table syntax. Tabel 1 shows an example.

Tabel 1. Example table

Parameter	Notation	Remarks
name	-	engine common identifier
manufacture	-	name of the manufacturer
bpr	λ	bypass ratio
pr	-	pressure ratio
thrust	T_0	maximum static thrust

6.4 Figures

Use standard Quarto figure syntax with `![caption] (path) {#fig-id width=X%}`. Place images in a `fig/` folder.

¹This is an example footnote.

6.5 Equations

Use standard LaTeX equation syntax. Persamaan 1 shows an example equation.

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \rho \mathbf{g} \quad (1)$$

7. Sections

Organize your paper using standard markdown headings.

Some standard sections are:

- Introduction
- Methods
- Results
- Discussion
- Conclusion

Acknowledgement

Include your acknowledgements in this section.

Funding Statement

When applicable, please specify the funding information for this research.

Reproducibility Statement

Mandatory section!

7.1 Information on how to reproduce this research, including access to: 1) source code related to the research, 2) source code for the figures, 3) source code/data for the tables when applicable.

8. Pendahuluan

9. Studi Literatur

XXXX

10. Metodologi

10.1 Data

10.2 Metode

11. Analisis